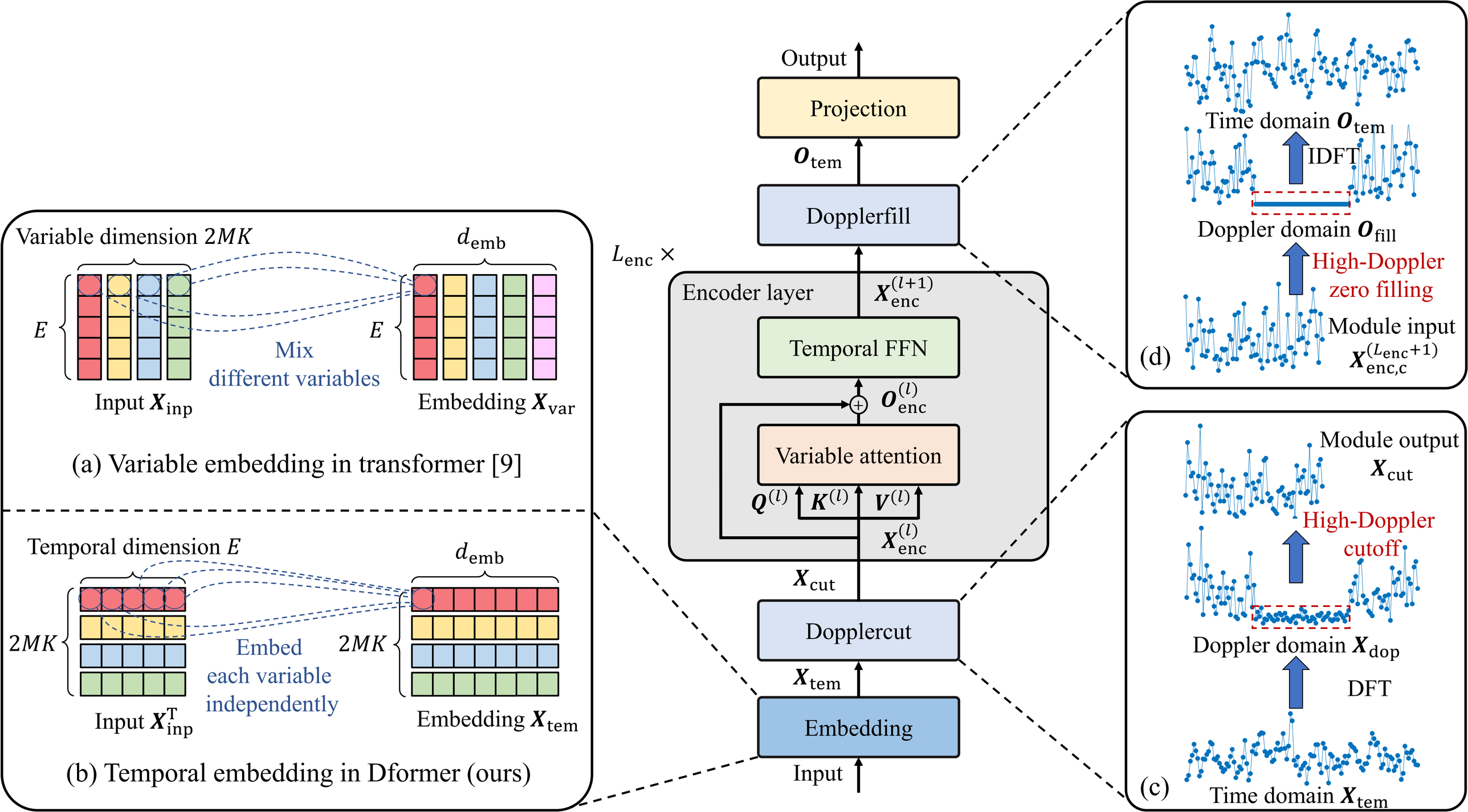

Variable dimension $2MK$
d_{emb}
E
E
Mix
different variables
Input $\boldsymbol{X}_{\text{inp}}$
Embedding $\boldsymbol{X}_{\text{var}}$
(a) Variable embedding in transformer [9]
Temporal dimension E
d_{emb}
$2MK$
$2MK$
Embed
each variable
independently
Input $\boldsymbol{X}_{\text{inp}}^{\text{T}}$
Embedding $\boldsymbol{X}_{\text{tem}}$
(b) Temporal embedding in Dformer (ours)
Output
Projection
$\boldsymbol{O}_{\text{tem}}$
Dopplerfill
$L_{\text{enc}}\times$
Encoder layer
$\boldsymbol{X}_{\text{enc}}^{(l+1)}$
Temporal FFN
$\boldsymbol{O}_{\text{enc}}^{(l)}$
Variable attention
$\boldsymbol{Q}^{(l)}$ $\boldsymbol{K}^{(l)}$ $\boldsymbol{V}^{(l)}$
$\boldsymbol{X}_{\text{enc}}^{(l)}$
$\boldsymbol{X}_{\text{cut}}$
Dopplercut
$\boldsymbol{X}_{\text{tem}}$
Embedding
Input
Time domain $\boldsymbol{O}_{\text{tem}}$
IDFT
Doppler domain $\boldsymbol{O}_{\text{fill}}$
High-Doppler
zero filling
Module input
$\boldsymbol{X}_{\text{enc,c}}^{(L_{\text{enc}}+1)}$
(d)
Module output
$\boldsymbol{X}_{\text{cut}}$
High-Doppler
cutoff
Doppler domain $\boldsymbol{X}_{\text{dop}}$
DFT
Time domain $\boldsymbol{X}_{\text{tem}}$
(c)